

Improving Knowledge Tracing with Contrastive Learning on Interaction Sequences

Abstract. Knowledge tracing (KT) models predict students' future interactions from their answering histories. We propose CLKT, an improved KT model that enhances the standard self-attention knowledge tracer with a supervised contrastive objective over interaction sequences. On three public datasets (ASSISTments, EdNet, XES3G5M), CLKT improves AUC by 2.1 points on average over the self-attention baseline while using the same input features.

1 Introduction

Knowledge tracing is the task of modeling a student's evolving knowledge state. The self-attention knowledge tracer treats interactions as a sequence but ignores the similarity structure between students' practice patterns. We argue that contrasting augmented views of the same interaction sequence yields better representations. Unlike prior KT work that adds external side information, our improvement requires no extra features and applies to any attention-based KT backbone.

2 Method

CLKT builds on the self-attention knowledge tracer. We construct two augmented views of each student's interaction sequence by masking and cropping, encode both views, and pull their representations together while pushing apart representations of different students (supervised contrastive loss). The contrastive loss is combined with the standard prediction loss with weight 0.5. No additional input features such as question difficulty or student metadata are required.

3 Results

On ASSISTments, CLKT reaches AUC 0.79 versus 0.77 for the self-attention baseline; on EdNet 0.74 versus 0.72; on XES3G5M 0.80 versus 0.78. The improvement is consistent across all three datasets. Ablation shows removing the supervised contrastive loss drops AUC back to baseline, confirming the gain comes from the contrastive objective. Training cost increases by only 15%.

4 Conclusion

A supervised contrastive objective over interaction sequences is a simple and effective way to improve attention-based knowledge tracing. Future work includes extending the contrastive scheme to graph-based KT models and studying cold-start students.